Project Design Phase-II
Technology Stack (Architecture & Stack)

Date	25 September 2026
Team ID	SWTID-2026-3161
Project Name	Script-Controlled ACL – Restrict Record Access Based on Field Value
Maximum Marks	4 Marks

Technical Architecture

The architecture below shows how a record access request flows through the Script-Controlled ACL system, from the user's request to the field-value evaluation, the access decision, and the audit trail an administrator can review.

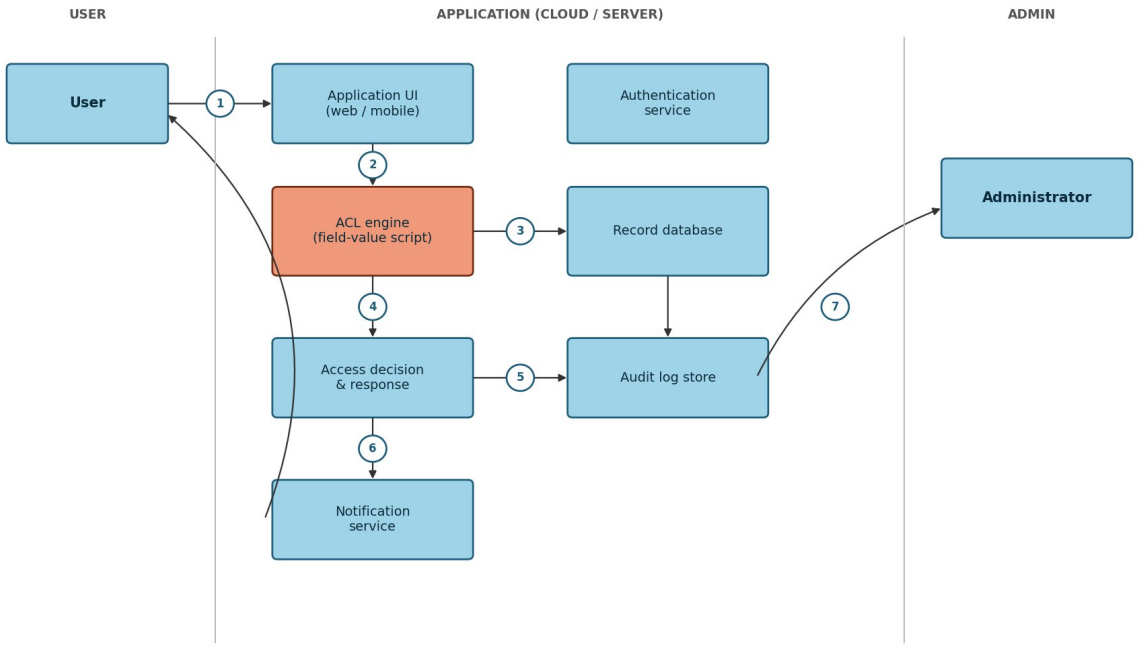


Table-1: Components & Technologies

S.No	Component	Description	Technology
1.	User interface	How the user requests record access	HTML, CSS, JavaScript / React JS
2.	Application logic-1	Authentication & request routing	Node.js / Java / Python
3.	Application logic-2	ACL engine – evaluates the field-value condition	Server-side script (Java / Python / JavaScript)
4.	Application logic-3	Access decision & response handling	Node.js / Java / Python
5.	Database	Stores the records the ACL protects	MySQL / PostgreSQL
6.	Cloud database	Database service on cloud	AWS RDS / IBM DB2 / MongoDB Atlas
7.	File storage	Audit log storage	AWS S3 or local filesystem
8.	External API-1	Identity / role lookup for the requesting	OAuth2 / LDAP API

S.No	Component	Description	Technology
		user	
9.	External API-2	Sends denial / alert notifications	SMTP / Twilio API
10.	Machine learning model	Not applicable to this feature	N/A
11.	Infrastructure (server / cloud)	Application deployment Local Server Configuration: Cloud Server Configuration:	Local, Cloud Foundry, Kubernetes, etc.

Table-2: Application Characteristics

S.No	Characteristics	Description	Technology
1.	Open-source frameworks	Frameworks used to build the ACL service and UI	Express.js / Spring Boot / Django
2.	Security implementations	Role check plus field-value ACL, encryption, and access controls	TLS/HTTPS, AES-256, IAM/OAuth2, OWASP practices
3.	Scalable architecture	Stateless ACL evaluation service that scales horizontally behind a load balancer	Microservice, container orchestration
4.	Availability	Multiple instances across zones with health checks and failover	Load balancer, multi-AZ deployment
5.	Performance	Caches frequently evaluated rules; logs audit entries asynchronously	In-memory cache, async queue

References

https://c4model.com/

https://www.ibm.com/cloud/architecture

https://aws.amazon.com/architecture

How to draw useful technical architecture diagrams